

Experimental and Prediction of Bubble Point Pressure for Binary and Ternary Systems Consisting of 1-Butanol, 2-Methyl-1-propanol, Glycerol and Water

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Table S1. Vapor pressures of 1-butanol and 2-methyl-1-propanol^a

1-butanol		2-methyl-1-propanol	
<i>T</i> /K	<i>P</i> /kPa	<i>T</i> /K	<i>P</i> /kPa
313.23	2.45	313.15	3.83
319.05	3.53	315.87	4.55
321.25	4.02	318.15	5.27
323.35	4.58	320.02	5.89
325.86	5.30	321.41	6.39
328.36	6.10	323.16	7.09
330.31	6.82	325.28	8.02
333.34	8.05	328.18	9.45
336.98	9.81	333.25	12.44
338.22	10.48		

340.91	12.02
343.26	13.54

^a Standard uncertainties u are $u(T) = 0.1$ K, and $u(P) = 0.1$ kPa.

Table S2. Parameters of the Antoine equation for pure compounds

component	A	B	C
1-butanol ^a	4.64930	1395.140	182.739
2-methyl-1-propanol ^a	4.34504	1190.380	166.670
glycerol ^b	10.6190	4487.040	-140.200
water ^a	5.11564	1687.537	230.170

^aref 27, $\log(P^s) = A - B/(T + C - 273.15)$, where P^s is in bar and T is in K.

^bref 28, $\ln(P^s) = A - B/(T + C)$, where P^s is in bar and T is in K.

Table S3. Physical properties and parameters of pure components in the activity coefficient correlation

component	molar volume/cm ³ ·mol ⁻¹	r^a	q^a
1-butanol	91.96 ^b	3.9243	3.668
2-methyl-1-propanol	92.91 ^b	3.9235	3.664
glycerol	73.03 ^c	4.7957	4.908
water	18.07 ^b	0.9200	1.400

^a ref 32. ^b ref 27. ^c ref 28.